Laboratory Project in Health Data Science

Objectives

The student is expected to develop a complete data science project applied to a real health problem, integrating:

- formulation of the research question/problem;
- critical use of Large Language Models (LLM);
- identification and selection of credible open data sources;
- data acquisition (preferably automated);
- appropriate statistical analysis;
- interpretation of statistical results;
- clear and visual communication of scientific information to non-statisticians;
- proper documentation.

Groups of 2 students (preferably) and a maximum of 3 students.

Project Stages

1) Definition of the research problem (with LLM)

The project must begin with the formulation of a clinical or public health research question.

- factors associated with cardiovascular mortality;
- vaccination trends;
- regional inequalities in health;
- environmental impacts on health.

This stage must be carried out with the support of an LLM (e.g., ChatGPT, Claude, etc.).

The group must:

- use the LLM to help define the question;
- refine the research question;
- justify its scientific relevance.

Mandatory submission (before the end of the project):

The GitHub repository must include a file named 'prompts.md' containing:

- all prompts used;

- LLM responses;
- critical comments by the students (the LLM can make mistakes).

The goal is not to have the LLM perform the work, but to demonstrate critical use.

2) Identification of an open data source

The dataset must be public and accessible online. With the support of an LLM, the group should:

- identify an open data source;
- plan variables;
- propose hypotheses;
- define an analysis strategy.

Possible sources (non-exclusive):

- Global Burden of Disease (IHME)
- National Statistics Institute (INE)
- PORDATA
- SNS Transparency Portal
- European Centre for Disease Prevention and Control (ECDC)
- Our World in Data
- Eurostat
- Ministry of Health Open Data
- WHO Global Health Observatory

Students must clearly indicate:

- source link;
- selected variables;
- time period analyzed;
- study population.

3) Data analysis

An appropriate statistical analysis must be performed and must be reproducible.

All code must:

- clean data;
- prepare variables;
- perform statistical analysis;
- automatically generate figures.

Accepted languages:

- Python

- R

Particularly valued:

- statistical models;
- inferential analysis;
- not only descriptive plots.

4) Interactive dashboard

Results must be presented through an interactive dashboard.

Possible tools (examples):

- Streamlit
- R Shiny
- Power BI

The dashboard must:

- answer the initial research question;
- allow user exploration;
- be understandable to non-specialists.

5) GitHub repository (main deliverable)

Each group must submit a repository containing:

- README.md (project explanation);
- prompts.md (interaction with the LLM);
- commented code;
- data (or download script);
- instructions to run the project;
- main results.

The project must be reproducible by another person.

6) Final presentation

Each group will give an oral presentation (project defense style):

- 10 minutes total;
- 5 minutes presentation;
- 5 minutes questions.

Teachers may ask about:

- methodological decisions;
- statistical analysis;

- use of the LLM;
- data limitations.

Students must demonstrate mastery of their work.

Assessment

Scientific quality of the research question — 10%

Data acquisition and processing — 20%

Statistical analysis — 25%

Dashboard — 20%

GitHub and reproducibility — 15%

Presentation and defense — 10%

Evaluation criteria

- coherence between question and analysis;
- critical thinking about the LLM;
- reproducibility;
- scientific clarity;
- visual communication;
- correct interpretation.